November 2026

Interpreting Hepatitis B Serological Laboratory Results

In Person Public Health Training

Justinian Wurtzel, MSc MPH
DrPH Student, Epidemiology



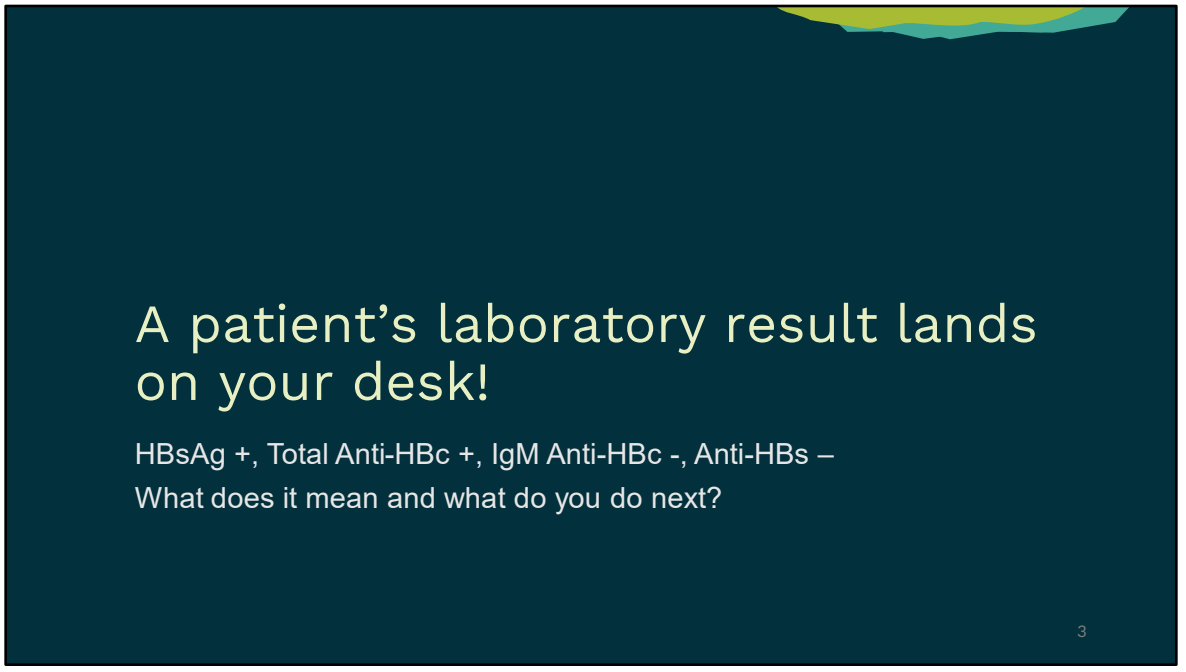
Title slide, ~time 5-10 minutes
Welcome and introductions.
Administer pre-test WITHOUT the handout, the participants are often clinical staff and can assess their own baseline understanding with the pre-test prior to the presentation
Pre knowledge check 10 questions
Best practices
Gagne's nine events of instruction: Stimulate recall of prior learning. Leverage students' existing knowledge before introducing new knowledge to scaffold learning.
Formative assessment: offering low risk assessment pretest
Privacy: only the participant and the facilitator know the initial scores
Transition After the assessments are passed they will be graded alongside the post-assessment to see difference in score.
Context:
This training assumes a basic level understanding of HBV

Learning Objectives

- By the end of the training the participants will be able to define key hepatitis B testing terms and serologic lab markers with at least 80% accuracy on the post knowledge check. Assessment part one quiz.
- Given a patient history and hepatitis B laboratory results participants will be able to interpret a patient's case classification and defend their assessment correctly for at least 2/3 patients. Assessment part two group interpretation and history.
- Given the patient history and lab result the participants will be able to classify a patient's case status using the 2024 CDC case definitions and handout for at least 2/3 patients. Assessment part two case classification.

2

Objectives, ~1-2 min
Which brings us to our learning objectives, we partially completed our first objective! By the end of the training the participants will be able to define key hepatitis B testing terms and serologic lab markers with at least 80% accuracy on the post knowledge check. Assessment part one quiz. (learning objective one)
Review three objectives and how they will be assessed, highlighting future activities.
Best practices
Gagne 2 Inform the students of the objectives. Establish the learning expectations for the course and how you will measure success or failure
Backward design, informing them of upcoming assessment, what they will need to do (to pay attention for during the training), and the metric for success
Transition
Ask group of participants who has interpreted HBV lab panels before?



The HOOK, ~2-3 min
Of those that raised their hands and those that may have knowledge, give them all a minute to process what these labs and results may mean and ask them "What do these labs mean? What can you tell me about this patient? What do you do next?" let one or two people answer and depending on the type of responses move into the transition
Best practices
Hook! Gagne 1: Gain the attention of the students. Tell a story or ask a thought-provoking question and "hook" the students with stimuli that catch and engage the brain to gain the students' attention.
Active learning
Universal design engagement
Transition
First to answer the question, we need a shared vocabulary and baseline understanding of terms and tests

HEPATITIS B

Key Definitions: Testing

- Screening Vs. Testing
  - Screening: testing people who are asymptomatic to find possible infection or immunity
  - Testing (diagnostic): testing people who are symptomatic, an identified exposure (e.g., sexual contact, injection drug use), or a prior positive result (e.g., HBsAg)
- Antigen Vs. Antibody
  - Antigen: a non-self molecule the immune system uses to identify pathogens (HBsAg, HBeAg)
  - Antibody: a protein the immune system makes to attack antigens-pathogens (Anti-HBs, Anti-HBe)

Notes, ~1-2 min
Knowledge sharing, screening versus testing and antigen versus antibody
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Transition
Next we will discuss some of the results of these test types

HEPATITIS B

Key Definitions: Results

- Qualitative Vs. Quantitative
  - Qualitative: positive/negative, non/reactive
  - Quantitative: a numeric value (Anti-HBs >=10 mIU/mL, HBV DNA 2,000 IU/mL)
- Detectable Vs. Undetectable
  - Detectable: at or above the lab test's limit of detection
  - Undetectable: below the lab test's limit of detection, not proof of zero (HBV DNA below cutoff is not positive but not negative)

Notes, ~1-2 min
Knowledge sharing, qual v quant and detectable v undetectable
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Transition
Next we will discuss some of the language used when discussing viral hepatitis

HEPATITIS B

Viral Hepatitis Language

- **Acute** (short-term/new) **infection:**
  - When the infection is newly acquired.
  - Approximately 80% of patients will be asymptomatic during the acute phase of infection.
- **Chronic** (long-term/previous) **infection:**
  - Infection lasts 6 months or more and is usually life-long unless treated.
- **Resolved or cleared infection:**
  - The body getting rid of the hepatitis infection—usually in the acute stage.
  - Patients can spontaneously clear the virus or clear the virus through treatment.

Notes ~1-2 min
Knowledge sharing, acute, chronic, resolved/cleared
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Transition
Next we will discuss the primary modes of transmission that we see in Iowa

HEPATITIS B

Modes of Transmission

- **Primary** (most common)
  - Perinatal; vertical transmission from a mother who has hepatitis B to her baby
  - Sharing injection drug use equipment
  - Sexual contact (e.g., anal, vaginal)
- **Secondary** (less common)
  - Needle-sticks; sharps injuries
  - Razors, glucose monitoring equipment, toothbrushes*
    - *If blood to blood transmission is present
  - Organ transplant and dialysis

Source: Centers for Disease Control and Prevention. Viral hepatitis surveillance—United States, 2017. Atlanta: US Department of Health and Human Services, Centers for Disease Control and Prevention; 2019. Available at: https://www.cdc.gov/hepatitis/statistics/2017surveillance/index.htm.

Notes, ~2-3 min
Knowledge sharing, primary and secondary modes of transmission
Discussing perinatal vertical transmission from a mother or birthing parent
Identifying people who use drugs NOT drug users or addicts
Highlighting perinatal transmission is most common in Iowa due to our increasing immigration and the "exposure" of immigration is not the cause for hepatitis B but reflective of the public health infrastructure to combat HBV in a person's home country
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Columbia 1: establish and support a class climate that fosters belonging for all students
Columbia 2: set explicit student expectations for an inclusive climate
Diverse populations
Transition
Next we will discuss why we test for HBV

HEPATITIS B

Why test for hepatitis B?

- Screening, one-time adult triple panel: HBsAg, Anti-HBs, Total Anti-HBc
- Occupational screening
- Diagnosis: acute/chronic, confirmed/probable
- Immune status: immune or susceptible
- Perinatal screen
- Monitoring chronic infection
- Exposure to contact with hepatitis B

Notes, ~1-2 min
Why do we test for HBV? The reason for the testing provides context on how you should read the result.
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Job relevance, occupational screening for participants, directly relevant to them
Transition
Next we will discuss serologic laboratory results

HEPATITIS B

Core Serologic Laboratory Results

- HBsAg, surface antigen
  - Viral hepatitis surface antigen, present in acute and chronic patients, infectious
- Total IgM/IgG Anti-HBc, core antibody
  - Past (resolved) or ongoing (chronic) infection, lifelong
- IgM Anti-HBc
  - Recent infection within six months, indicative of acute infection
- Anti-HBs, surface antibody
  - Recovery and immunity from infection or successful vaccination

Notes ~2-3 min
The core serology panel tests and results
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Question: WHICH serology markers can a vaccine never produce? And why? IgM Anti-HBc core antibody. The vaccine only contains surface antigen and will produce true or false positive, and will elicit immune response anti-HBs but not the core antibody
Formative assessment, short knowledge check to keep people engaged
Transition
Next we will discuss other useful serology labs

HEPATITIS B

Additional Labs

- HBeAg and Anti-HBe
  - E-antigen active viral replication and antibody to the e-antigen
- HBV DNA
  - Genetic viral material, quantitative results for viral load
- ALT, AST, Bilirubin
  - Liver damage ALT>200 IU/L or bilirubin>=3.0mg/dL meet acute definition

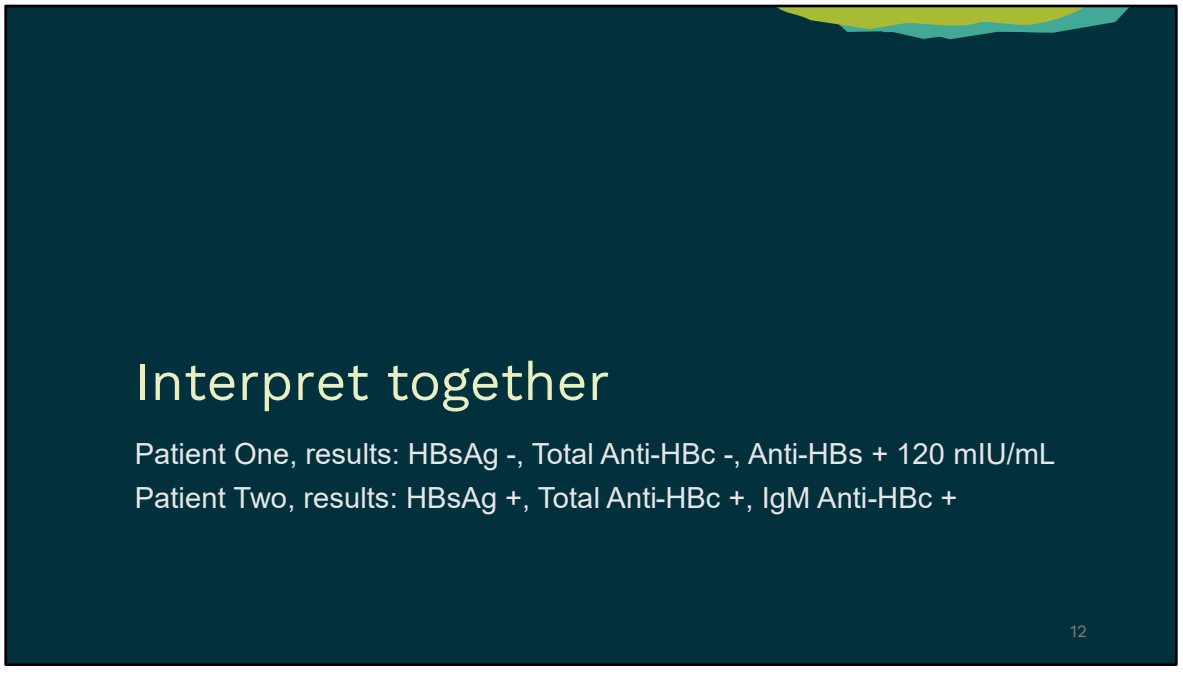
Notes, ~1-2 min
Other helpful serology panel tests and results and the liver function tests are significant for their importance in acute laboratory criteria
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Transition
Next we will discuss interpreting lab results

HEPATITIS B

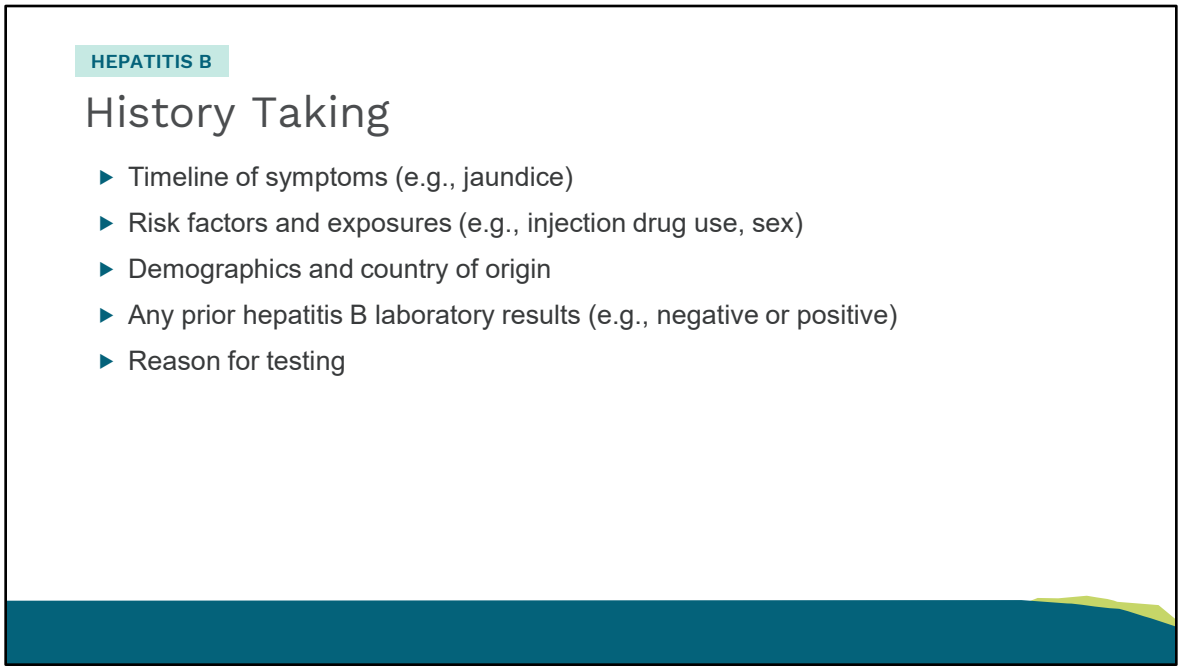
Interpreting panels

HBsAg	Total Anti-HBc	IgM Anti-HBc	Anti-HBs	Interpretation
-	-	-	-	Never infected, susceptible, vaccinate
+	-	-	-	Early acute or false positive up to 18 days after vaccination
+	+	+	-	Acute infection
-	+	+	-	Acute infection, resolving
-	+	-	+	Recovered from past infection, immune
+	+	-	-	Chronic infection
-	+	-	-	Isolated core, past infection, low level chronic, or false positive
-	-	-	+	Immune from vaccination (Anti-HBs>=10mIU/mL)

Notes, ~5 min
Review interpretations line by line with participants
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Gagne 7: Provide feedback. Reinforce knowledge with immediate feedback.
Return of the HOOK (slide 3)
“What do these labs mean? What can you tell me about this patient? What do you do next?” let one or two people answer and depending on the type of responses move into the transition
Chronic confirmed hepatitis B patient
Transition
Next we will discuss interpreting two more patient results together



Notes ~5 min
Allow participants to review patient results one and two and wait for their responses
Patient one, vaccinated and immune
Patient two, acutely infected within the last six months
Best practices
Active learning
Gagne 5: Provide learner guidance. Use examples, case studies, stories, discussions, and other instructional support materials to guide students' learning.
Transition
Next we will discuss important aspects of history taking in developing serological context



Notes ~1-3 min
Review important items in hepatitis history taking
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Real aspects of people's lives, the investigation and interview allow for connection building with the patient and help identify more medically relevant information
Transition
Next we will discuss notable circumstances

HEPATITIS B

Notable Circumstances

- False positive HBsAg due to recent hepatitis B vaccination (most common)
- Perinatal hepatitis B and infants need HBIG if mother is hepatitis B positive
- Immunocompromised patients
- Reactivation events in chronic or resolved infection during immunosuppressive therapies.

Notes ~1-3 min
Highlight the notable circumstances, important to highlight these unique circumstances as they may obfuscate testing results (a false positive) or reactivating immunocompromised patient (chronic not acute)
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Real aspects of people's lives, the investigation and interview allow for connection building with the patient and help identify more medically relevant information
Transition Next we will discuss common mistakes we all make

HEPATITIS B

Common Mistakes

- A single positive result sent to state, negatives not shared (follow-up required)
- Disagreeing results (e.g., HBsAg – but HBV DNA +)
- IgM Anti-HBc in a reactivating chronic patient may read acute without additional lab results and history
- Unconfirmed laboratory results (only positive labs are reportable)
- Low or equivocal results
- No vaccination record

Notes ~1-3 min
Review common serological mistakes
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Peer learning, mistakes public health staff and other providers have made
Transition
Next we will review the 2024 CDC Case Definitions

HEPATITIS B

CDC 2024 Case Definitions

Classification	Laboratory Evidence	Clinical Criteria
Acute, Confirmed Tier 1	Positive HBsAg, HBeAg, or HBV DNA with IgM Anti-HBc or seroconversion within 12 months of negative HBsAg	No
Acute, Confirmed Tier 2	Positive HBsAg or HBV DNA with IgM Anti-HBc not done	Yes
Acute, Probable	Positive IgM Anti-HBc with HBsAg, HBeAg and HBV DNA negative or not done	Yes
Chronic, Confirmed	Positive HBsAg/HBeAg twice six months apart or HBsAg/HBeAg plus Total IgM/IgG Anti-HBc or HBsAg plus HBeAg or HBV DNA	No
Chronic, Probable	HBsAg or HBeAg with negative IgM Anti-HBc or not done	No

Notes ~5 min
Review the 2024 CDC Case Definitions
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Comparison chart visualization
Universal design, representation, building knowledge and language and symbols
Transition
After interpretation, classify the patient

HEPATITIS B

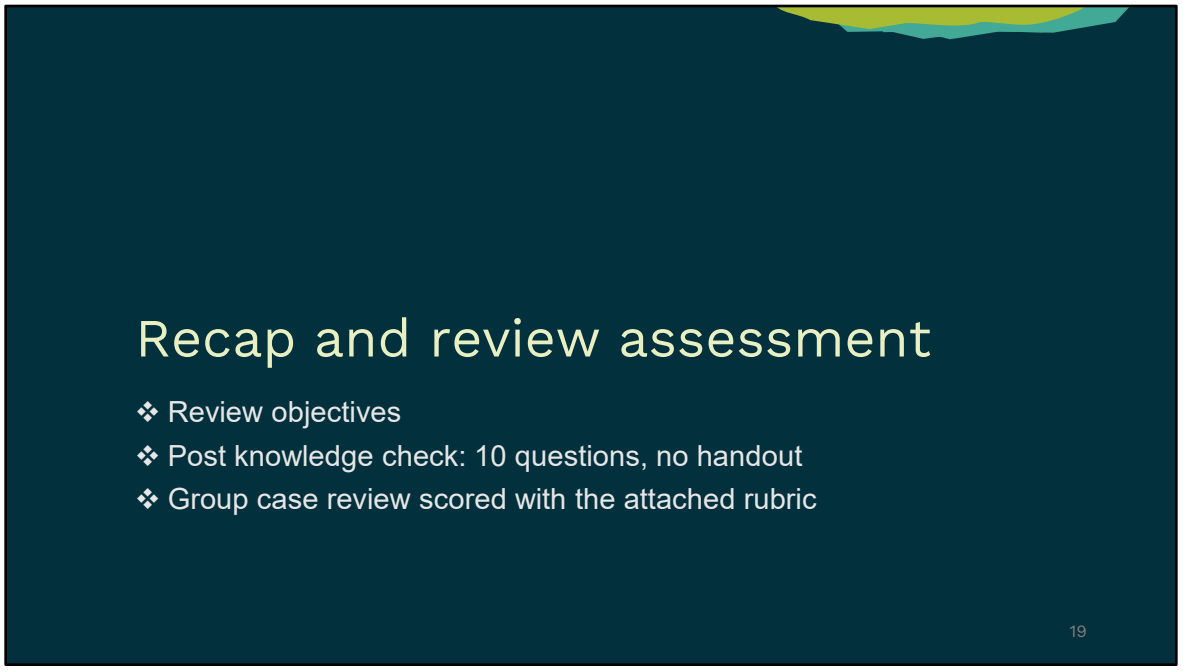
After Interpretation, Classification

- Linkage to care or ensure primary care provider
- Vaccinate any susceptible patients
- Contact tracing: any household, sexual, or injection drug use contacts
- Patient and provider follow-up

Notes ~2-3 min
After interpretation we classify and begin next steps, activities the provider or staff member can do once the patient is classified
Best practices
Universal design representation: building knowledge
Gagne 4: Present the content. Deliver the content in smaller consumable chunks for easier information processing.
Gagne 9: Enhance retention and transfer to the job. Use content retention strategies to retain knowledge to include but not limited to concept mapping, rephrasing, summarizing, and job aids
Transition
Group Case Review



Notes ~30 min
Group case review, see assessment part two for group assignment, you can use handout!
Best practices
Practice based learning
Universal design action and expression: interaction, expression & communications
Gagne 6: Elicit performance. Engage students with activities that help them recall, utilize, and evaluate knowledge.
Gagne 9: Enhance retention and transfer to the job. Use content retention strategies to retain knowledge to include but not limited to concept mapping, rephrasing, summarizing, and job aids
Transition
Recap and review assessment



Notes ~5 min recap, ~10 min assessment, no handout, assessments will be reviewed during this time, recorded, and returned to participants.
Best practices
Practice based learning
Universal design action and expression: interaction, expression & communications
Gagne 2: Inform the students of the objectives. Establish the learning expectations for the course and how you will measure success or failure
Gagne 6: Elicit performance. Engage students with activities that help them recall, utilize, and evaluate knowledge.
Gagne 7: Provide feedback. Reinforce knowledge with immediate feedback.
Gagne 8: Assess performance. Test the learner's knowledge and skills.
Gagne 9: Enhance retention and transfer to the job. Use content retention strategies to retain knowledge to include but not limited to concept mapping, rephrasing, summarizing, and job aids
Transition
Thank everyone for their time, record final post assessment results and return the pre-post, and group assessment documentation for their records.